

Non-LLM Complexity Addendum

April 7, 2026

Scope

This addendum extends the approved-solver complexity analysis to the non-LLM systems stored in this repository.

Data

- ARC-1 approved eval overlap: 38 tasks
- ARC-2 approved eval overlap: 18 tasks
- ARC-2 tasks with both human and non-LLM difficulty: 17 tasks
- ARC-1 non-LLM profiles: 11
- ARC-2 non-LLM profiles: 28

Methods

For each dataset I built a task-by-profile success matrix and derived smoothed-logit, PCA, and penalized 1PL item-difficulty summaries. The key-test table uses the smoothed-logit difficulty as the primary outcome. Statistical inference used bootstrap confidence intervals, permutation p-values for direct correlations, bootstrap difference-of-correlation tests, and BH-FDR correction across the 9 key hypotheses.

Headline Results

Best ARC-2 structural metric: Gzip bytes ($r = 0.504$, $q = 0.217$).

Best ARC-1 structural metric: Gzip bytes ($r = 0.366$, $q = 0.191$).

Best pooled metric after within-dataset standardization: Gzip bytes ($r = 0.378$, $q = 0.062$).

ID	Estimate	CI low	CI high	p	q	FDR
N1	0.113	-0.364	0.518	0.657	0.739	no
N2	0.062	-0.381	0.440	0.818	0.818	no
N3	0.469	0.109	0.864	0.518	0.721	no
N4	0.352	-0.157	0.660	0.148	0.667	no
N5	0.167	-0.170	0.742	0.561	0.721	no
N6	0.346	-0.044	0.658	0.508	0.721	no
N7	0.356	-0.066	0.817	0.495	0.721	no
N8	0.255	-0.042	0.507	0.125	0.667	no
N9	0.113	-0.211	0.399	0.499	0.721	no

Interpretation

The cleanest read from this first pass is that non-LLM item difficulty is not random with respect to approved solver complexity, but the structural-complexity link is weaker and less focused than the LLM-side result. The ARC-2 non-LLM profiles still line up more naturally with the LLM difficulty axis than with the human axis, and once LLM difficulty is controlled the remaining structural signal in the non-LLM axis is smaller and less secure. A pooled within-dataset-standardized view strengthens the directional pattern, which suggests that low power is part of the story in the split analyses.

Non-LLM Complexity Pass: ARC-2 Relationships

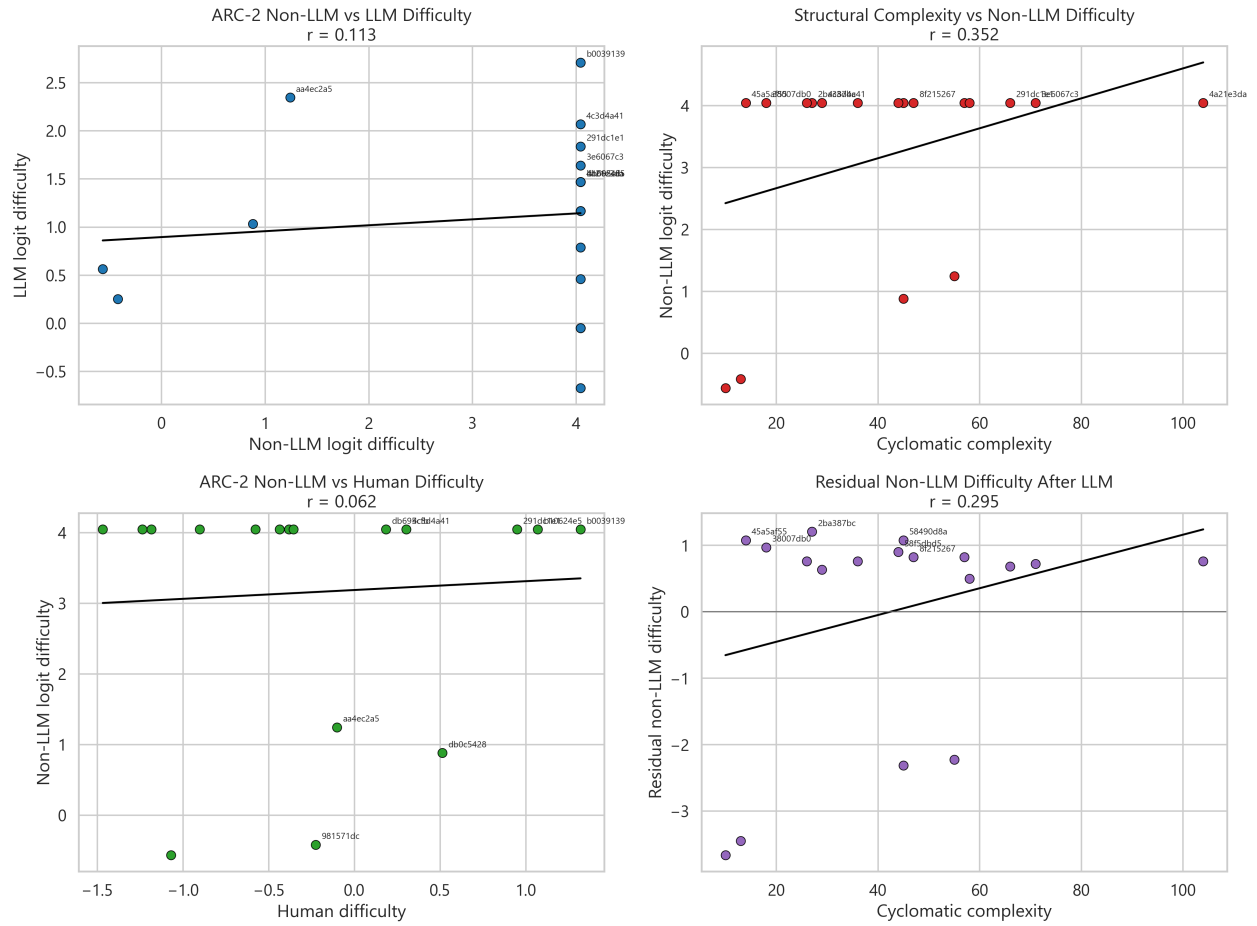


Figure 1: ARC-2 non-LLM relationships.

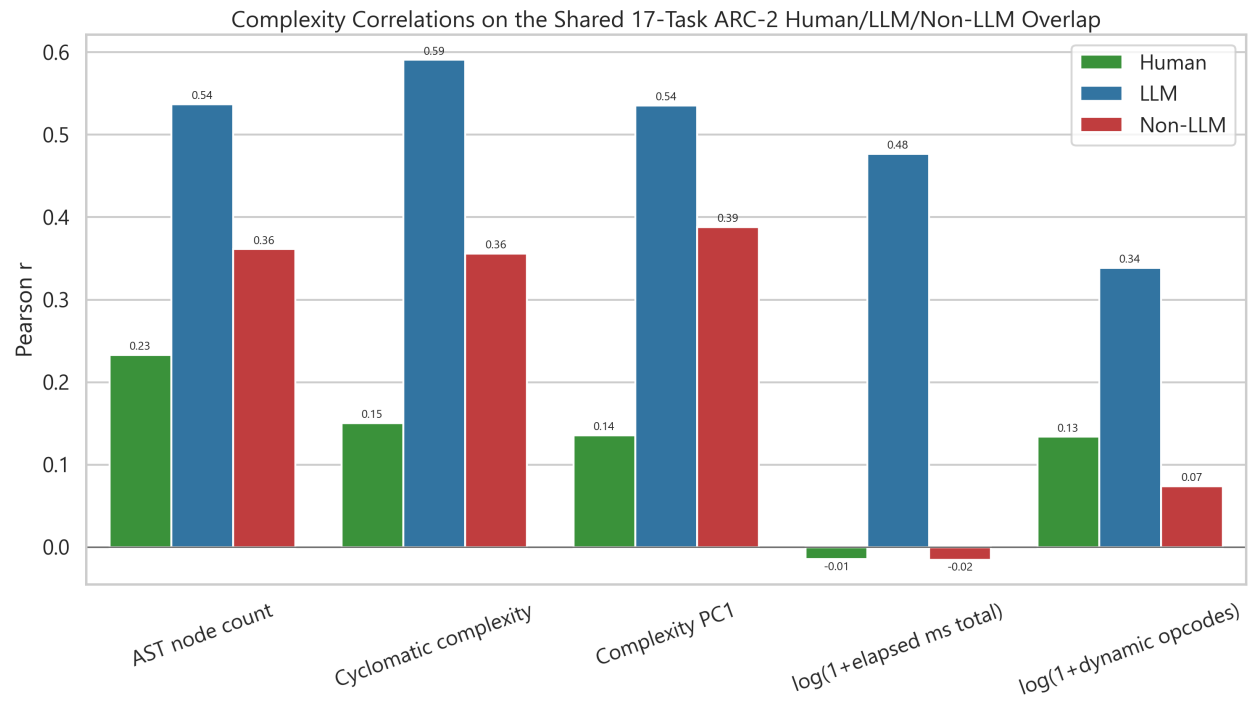


Figure 2: Human vs LLM vs non-LLM complexity comparisons on the shared 17-task ARC-2 overlap.

Non-LLM Complexity Pass: ARC-1 Sidecar

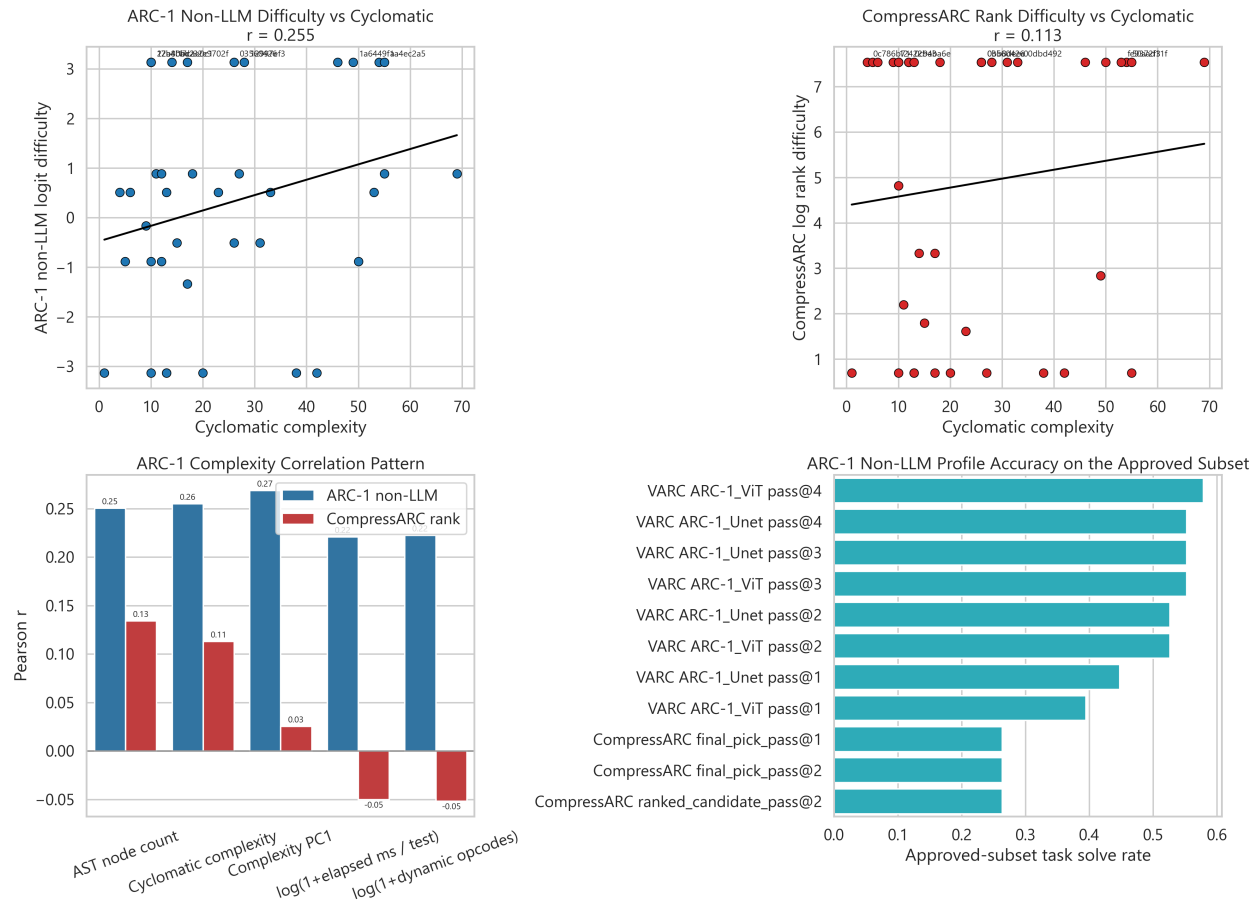


Figure 3: ARC-1 sidecar results.

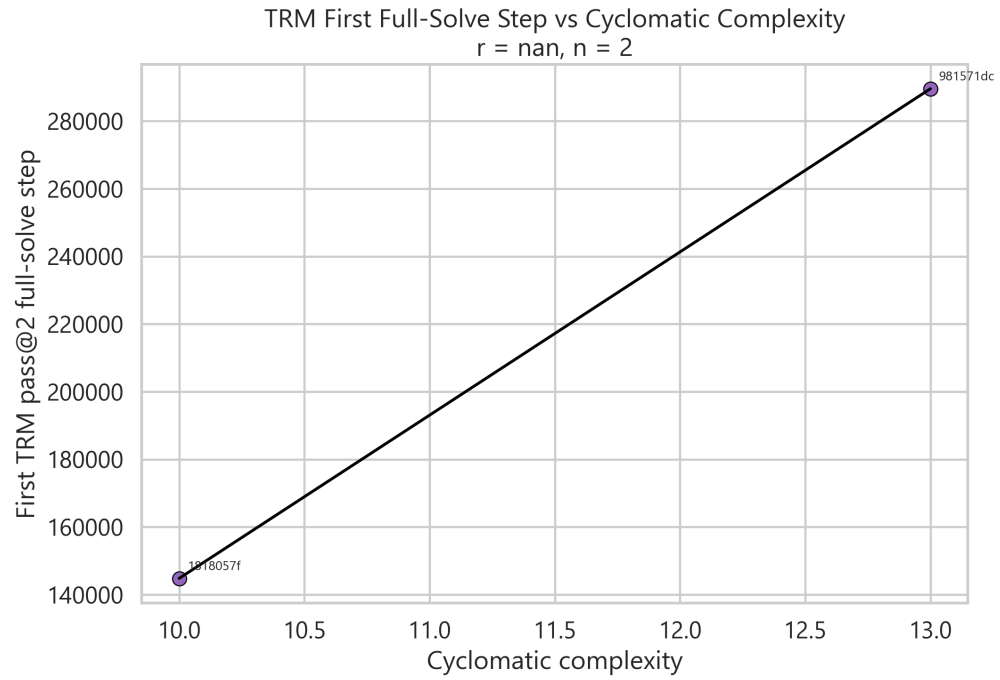


Figure 4: Exploratory TRM trajectory vs complexity.